PostgreSQL-Backed Transport Analytics for Paris and New York City

Big Data Project

April 13, 2026

Abstract

This project analyzes public-transport demand in Paris and New York City using PostgreSQL as the single analytical source of truth. Source tables in the `public` schema are exposed through reporting-clean views in the `transport` schema, then consumed by a Python workflow that generates reproducible result tables and figures. The final method set contains four components: temporal profiling, lag-based forecasting, anomaly detection, and contributor / city-structure comparison. On the current reporting scope, the forecasting baseline achieves an overall mean absolute percentage error of 6.02%, with stronger performance for NYC (5.86%) than Paris (6.94%). Paris shows a higher anomaly rate (8.96%) than NYC (0.73%), while Paris also exhibits higher average daily demand ($\approx$ 4.13 million) than NYC ($\approx$ 2.63 million).

1 Introduction

The goal of this project is to compare large-scale transport demand patterns in Paris and New York City using one reproducible analytical workflow. Earlier iterations of the repository mixed exploratory notebook material, local-file assumptions, and placeholder narrative. The current baseline removes that ambiguity: PostgreSQL stores the authoritative source data, SQL views define the reporting contract, and Python produces the final derived outputs under `report/results/`.

This report is written from the current generated outputs rather than from earlier placeholder material. The emphasis is on defensible interpretation: how demand evolves over time, how well a simple forecasting model performs, which observations are anomalous, and which stations dominate total traffic.

2 Data and Architecture

The project reads three source datasets already loaded into PostgreSQL:

- `public.idfm_daily_validations` with station-level Paris daily validations,
- `public.idfm_hourly_profiles` with Paris hourly validation shares,
- `public.mta_hourly_ridership` with NYC hourly ridership.

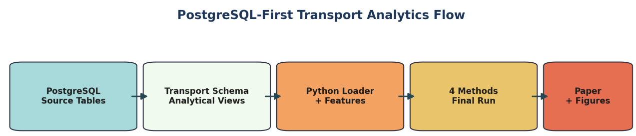

Figure 1: Final PostgreSQL-first analysis flow used by the project.

Reference tables `public.idfm` and `public.mta` provide network metadata. The analytical contract is defined in the `transport` schema:

- `v_daily_demand` for reporting-clean daily demand,
- `v_nyc_hourly_profile` and `v_paris_hourly_profile`,
- `v_contributor_source` for station contribution ranking,
- `v_city_structure_source` for city-level comparison.

Incomplete boundary years are excluded from the reporting views, which removes the partial Paris 2025 year from the final result set. The final workflow contains 12,696 daily rows and 2,003 contributor rows in the generated summary output.

3 Methods

The final method set contains four components.

Temporal profiling. Daily demand is summarized by weekday, month, and year, while hourly profiles are analyzed separately for NYC ridership and Paris validation shares.

Forecasting. A lag-based random forest baseline uses lag features, rolling statistics, and calendar features. The model is evaluated with a time-ordered train/test split.

Anomaly detection. A hybrid rule combines rolling z-scores with an `IsolationForest` model fitted on value and lag-derived features.

Contributor and city-structure analysis. Station-level demand is ranked to identify dominant contributors, and city-level summaries are computed for average daily demand, variability, seasonality, and weekend sensitivity.

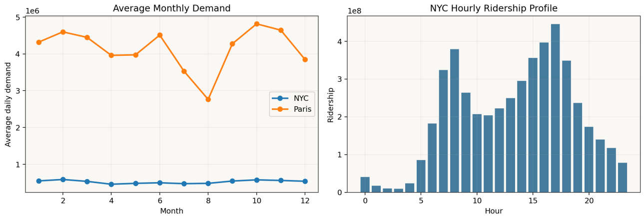

Figure 2: Monthly demand and NYC hourly ridership profile from the current run.

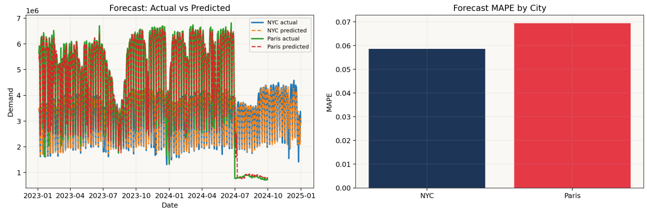

Figure 3: Forecast quality from the current PostgreSQL-backed result set.

Table 1: Forecast accuracy by city

City	MAE	RMSE	MAPE
NYC	29,402	62,965	5.86%
Paris	187,446	376,361	6.94%

Table 2: City structure summary

City	Avg daily	Weekend/weekday	Peak month
NYC	2.63M	0.584	2
Paris	4.13M	0.512	10

4 Results

4.1 Temporal patterns

The yearly totals confirm the reporting-clean scope: NYC covers 2020–2024 and Paris covers 2015–2024, with the partial 2025 year removed. Monthly structure is visibly different across the two systems. NYC peaks in winter and early autumn, while Paris peaks in October and weakens sharply in August.

4.2 Forecasting

The forecasting baseline remains strong enough to support the final paper. The overall model is trained on 8,336 rows and evaluated on 4,276 rows, with an overall MAE of 52,909, RMSE of 156,343, and MAPE of 6.02%.

NYC forecasts are more stable in absolute terms, but both cities remain within a single-digit MAPE range. This is sufficient for a baseline forecasting section in the final report, although the Paris series is clearly harder to predict due to larger swings and lower short-term regularity.

4.3 Anomalies

Anomaly rates remain highly asymmetric between the two cities. NYC has an anomaly rate of 0.73%, while Paris reaches 8.96%. The largest Paris anomalies cluster around 2020 disruption and recovery periods, whereas NYC outliers are concentrated around holiday and unusually weak weekend traffic.

4.4 Contributors and city structure

Paris has the larger system-level demand in the current reporting scope. Its average daily demand is approximately 4.13 million versus 2.63 million for NYC. Paris also shows a lower weekend-to-weekday ratio (0.512) than NYC (0.584), which indicates stronger weekday concentration.

The top contributor table is plausible after label cleanup. Paris is dominated by Saint-Lazare, La Défense–Grande Arche, Gare de Lyon, Gare du Nord, and Montparnasse. NYC is led by Times Sq–42 St, Grand Central–42 St, 34 St–Herald Sq, and 14 St–Union Sq.

5 Discussion and Limitations

The project is now aligned around a reproducible database-backed workflow, but several limitations remain.

First, the forecasting model is intentionally a baseline rather than a final production forecaster. It captures broad structure well, but it does not model special events, strikes, weather, or network interventions.

Second, the anomaly analysis is descriptive rather than causal. The flagged observations indicate unusual behavior, but they do not prove why it happened.

Third, contributor ranking depends on the quality of station labeling in the source tables. The obvious whitespace issues were removed during Stage 2 cleanup, but semantically similar stations are still treated as distinct when their identifiers differ.

Finally, the notebook and report are derived from generated outputs rather than from live database queries at compile time. That is acceptable for submission, but it means the result regeneration step remains part of the reproducibility contract.

6 Conclusion

The project now has a coherent final analytical baseline. PostgreSQL is the single source of truth, the reporting views remove incomplete comparison years, and the generated outputs support a defensible paper narrative. The main findings are consistent across the final result set: Paris has higher overall demand, stronger weekday concentration, and a much higher anomaly rate, while NYC is easier to forecast with the current lag-based baseline. The repository is now ready for final notebook alignment and submission cleanup rather than another major analytical redesign.

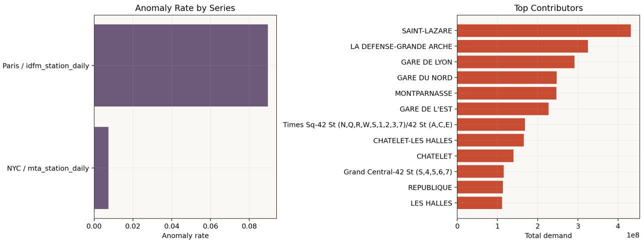

Figure 4: Anomaly-rate contrast and top contributors in the cleaned result set.

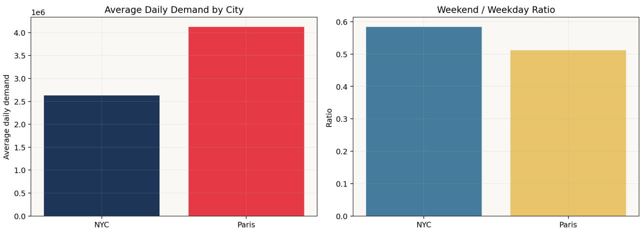

Figure 5: Average daily demand and weekend sensitivity by city.

References

1. Project background papers stored under `docs/papers/`.
2. Generated analytical outputs stored under `report/results/`.
3. PostgreSQL analytical contract defined in `sql/02_views.sql`.